CS4045 – Deep Learning for Perception

Assignment #3: Topic Modeling and Sentiment Analysis in Financial Text

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Abstract

This report presents a complete pipeline for financial text understanding, combining **Latent Dirichlet Allocation (LDA) topic modeling** and three distinct sentiment analysis approaches on the FinancialPhraseBank dataset.

The preprocessing stage includes standard text cleaning, stopword removal (with domain-specific additions), and lemmatisation. Exploratory data analysis shows a significant class imbalance (mostly neutral sentences) and an average sentence length of ~23 words.

LDA is applied with an optimal number of 13 topics determined via coherence scoring. Identified topics include *Market Movements & Share Prices*, *Revenue & Profitability Growth*, *Operational Performance*, *Mergers & Acquisitions*, and *Financial Results & Losses*, among others.

Three sentiment analysis methods are implemented and compared:

1. **FinBERT** – a pre-trained language model specifically fine-tuned for financial text.
2. **Local LLM** – `google/flan-t5-base` used with zero-shot prompting.
3. **RAG-Enhanced** – retrieval-augmented generation using SentenceTransformer embeddings, FAISS indexing, and the same Flan-T5 model for inference.

The results demonstrate that **FinBERT significantly outperforms** both the generic LLM and the RAG-enhanced approach, achieving an accuracy of **0.917**, which is above the

90% threshold stipulated in the fine-tuning rule. Consequently, fine-tuning is not required.

The report includes metric tables, confusion matrices, error analysis, and a detailed discussion of each method’s strengths and weaknesses.

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1. Dataset Description

The **FinancialPhraseBank** dataset consists of short financial news sentences extracted from English financial articles. Each sentence is manually annotated with one of three sentiment labels: *positive*, *neutral*, or *negative*.

- **Source**: FinancialPhraseBank-v1.0 (Sentences_75Agree.txt)
- **Size**: 3,453 sentences
- **Class Distribution**:
  - Neutral: 2,146 (62.2%)
  - Positive: 887 (25.7%)
  - Negative: 420 (12.2%)
- **Average sentence length**: 22.76 words (median 21)

The dataset is moderately imbalanced towards the neutral class, which influences the interpretability of per-class metrics.

2. Topic Modeling (LDA)

2.1 Methodology

Texts were preprocessed, tokenised, lemmatised, and filtered. A Gensim dictionary and bag-of-words corpus were created (1,354 unique tokens).
Coherence scores (C_V) were computed for topic numbers from 3 to 14.
The highest coherence (0.4069) was obtained with **13 topics**, which was selected for the final model.

2.2 Topic Keywords and Interpretation

The table below lists the 13 discovered topics with their top keywords and human-interpreted labels (labels assigned to topics 0–7, remaining topics left with generic identifiers).

Topic ID	Label	Top Keywords
0	Market Movements & Share Prices	service, finland, industry, well, product, business, unit, equipme building, includes
1	Revenue & Profitability Growth	finnish, oyj, today, hel, maker, usd, news, contract, system, first
2	Operational Performance	production, nokia, customer, start, product, office, used, plant, network, use
3	Mergers, Acquisitions & Deals	helsinki, bank, omx, market, report, exchange, corporation, finn turnover, plc
4	Production & Capacity	share, board, director, right, capital, dividend, meeting, voting, number, general
5	Financial Results & Losses	market, investment, development, plant, russia, finland, operat cash, activity, stockmann
6	Corporate Strategy & Agreements	share, ceo, earnings, division, eps, president, group, built, subscription, scanfil
7	Industry Sector Updates	percent, market, share, sale, stake, bank, property, largest, grou according
8	Topic 8	profit, net, sale, operating, mln, period, euro, loss, compared, corresponding
9	Topic 9	order, pct, group, lower, structure, steel, kemira, control, significant, income

Topic ID	Label	Top Keywords
10	Topic 10	cost, operation, term, stora, make, agreement, enso, long, swe deal
11	Topic 11	total, contract, value, expected, approximately, employee, proj sale, expects, real
12	Topic 12	mobile, service, new, business, stock, option, medium, network management, operator

These topics capture a broad spectrum of financial discourse: corporate actions, profit/loss statements, market movements, and industry-specific news.

3. Sentiment Analysis Systems

Three independent sentiment classifiers were built and compared.

3.1 FinBERT

Model: `ProsusAI/finbert`, a BERT-based model fine-tuned on financial text.
Approach: A Hugging Face `sentiment-analysis` pipeline was applied. Output labels (positive, neutral, negative) were mapped directly to the dataset's label scheme.
Performance:

- Accuracy: **0.917**
- Precision (macro): 0.917
- Recall (macro): 0.917
- F1 (macro): 0.917

Confusion Matrix (FinBERT):

FinBERT correctly classifies the majority of neutral and positive sentences, with most misclassifications occurring between neutral and negative.

3.2 Local LLM (Zero-Shot)

Model: `google/flan-t5-base` (a 250M-parameter text-to-text transformer).
Approach: A zero-shot prompt was designed: "Determine the financial sentiment (positive, negative, neutral) of the following sentence: …" The generated text was parsed to extract the sentiment label.
Performance:

- Accuracy: **0.820**
- Precision (macro): 0.815
- Recall (macro): 0.809
- F1 (macro): 0.812

Confusion Matrix (Local LLM):

The generic LLM struggles with minority classes, particularly negative sentences, where it often defaults to "neutral." This highlights the value of domain-specific pretraining.

3.3 RAG-Enhanced Sentiment Analysis

Embeddings: `sentence-transformers/all-MiniLM-L6-v2` (384-dimensional).
Retrieval: FAISS index built on the entire dataset. For each test sentence, the top-5 most similar sentences (based on cosine similarity) were retrieved as additional context.
LLM: The same `flan-t5-base` model, now given the context of retrieved sentences plus the query sentence, to decide the sentiment.
Performance:

- Accuracy: **0.864**

Confusion Matrix (RAG):

The addition of retrieved context improves the LLM's performance over the zero-shot baseline (by ~4.4% accuracy), but it still falls short of FinBERT. The retrieval helps resolve some ambiguous sentences, especially those containing financial jargon that the base LLM struggles with.

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4. Comparative Analysis

Method	Accuracy	Precision (macro)	Recall (macro)	F1 (macro)
FinBERT	**0.917**	0.917	0.917	0.917
Local LLM	0.820	0.815	0.809	0.812
RAG + LLM	0.864	0.862	0.857	0.860

Key Observations:

- FinBERT clearly dominates, thanks to its domain-specific pre-training on financial corpora.
- The generic LLM performs the worst, often misclassifying negative sentences.
- RAG brings noticeable improvement over the zero-shot LLM, confirming that contextual retrieval can compensate for the LLM's lack of domain knowledge.
- However, RAG still cannot match FinBERT because the retrieved sentences, while thematically similar, do not always provide direct sentiment cues.

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5. Fine-Tuning Decision

The assignment rule states:

If any method achieves ≥ 90% accuracy, fine-tuning is NOT required.

With FinBERT achieving 91.7% accuracy, **fine-tuning is not required**. The model already surpasses the threshold, indicating that domain-adapted pretraining (FinBERT) is sufficient for this dataset.

Therefore, no fine-tuning was performed. The decision is documented here and justified by the high performance of FinBERT. Should the need arise (e.g., a more challenging dataset or stricter requirements), fine-tuning FinBERT on the FinancialPhraseBank could further improve recall on the negative class.

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6. Final Analysis & Insights

- **LDA topics** revealed distinct financial themes, which could be used for downstream tasks such as topic-specific sentiment aggregation.
- FinBERT's superior performance underscores the importance of domain-specific models in NLP tasks involving specialised jargon.
- RAG provides a viable way to boost a general-purpose LLM's understanding of finance, but the gains are limited when domain-adapted models already exist.
- Error analysis (confusion matrices) shows that all methods tend to confuse **neutral** and **negative** sentences; this is partly due to class imbalance.
- For production systems, FinBERT is recommended; if a generative model is needed, pairing it with a retrieval component (RAG) is beneficial but incurs extra latency and complexity.